Crush Cup : Revival of the Terracotta

Masters Degree Thesis Project - GSA, Scotland, UK | Creative Enterprise Awardee 2023 (CEA) Prize Winner

Click to read thesis pdf in detail

Project Background - Stage 1 :

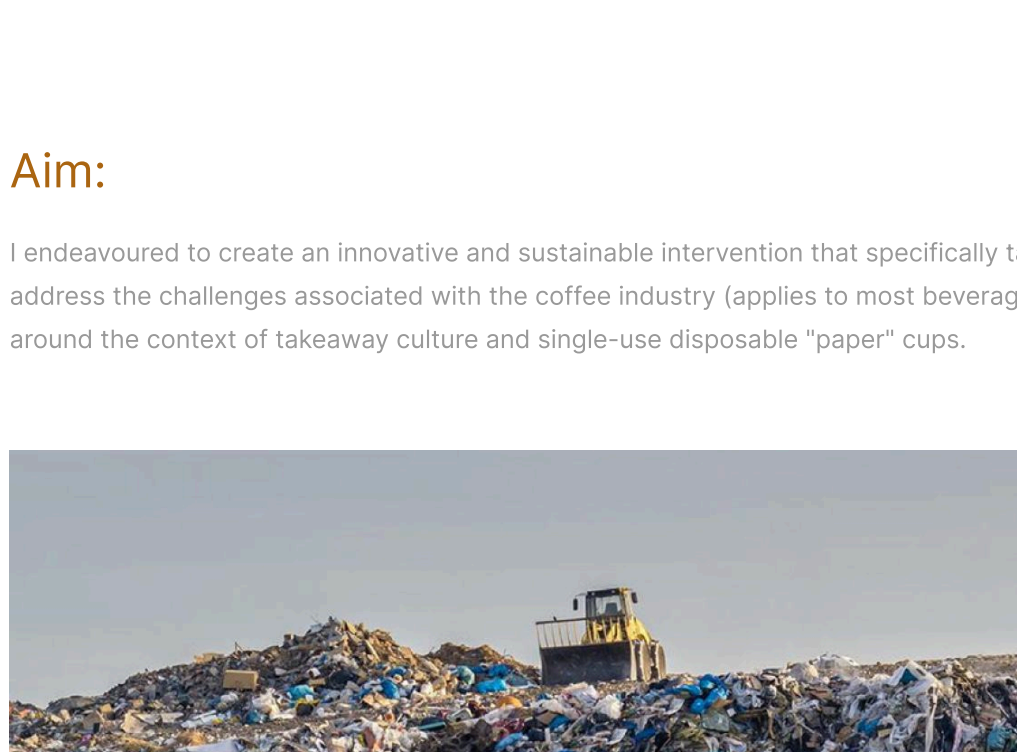
The UK faces a pressing issue with over 2.5 billion disposable coffee cups discarded annually, with limited recycling. These cups, made of paper and plastic, are challenging to recycle due to waterproof coatings. Their disposal contributes to landfill waste, emitting potent greenhouse gas methane. This situation offers research opportunities in areas like public perceptions, coffee consumption practices, and the takeaway coffee culture. Drawing inspiration from India's traditional tea culture, specifically the kullads and Bhar cups, this proposal aims to establish a sustainable service paradigm for daily coffee consumers. The use of terracotta cups as an alternative to disposable coffee cups.

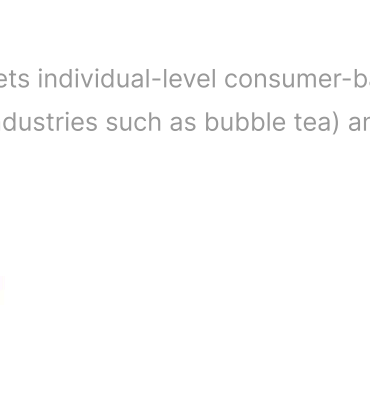
THE GLASGOW SCHOOL OF ART

TANVI KUMAR | M.DES SERVICE DESIGN | 2022–23

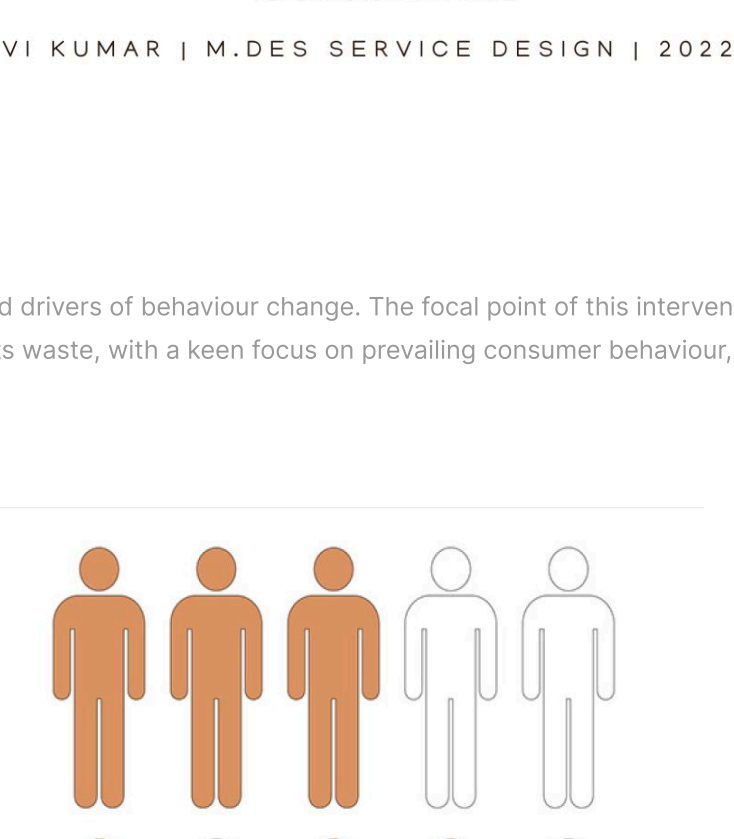
Aim:

I endeavoured to create an innovative and sustainable intervention that specifically targets individual-level consumer-based drivers of behaviour change. The focal point of this intervention will be to address the challenges associated with the coffee industry (applies to most beverage industries such as bubble tea) and its waste, with a keen focus on prevailing consumer behaviour, especially around the context of takeaway culture and single-use disposable "paper" cups.





2.5 billion disposable cups are disposed each year, only 1% is recycled.



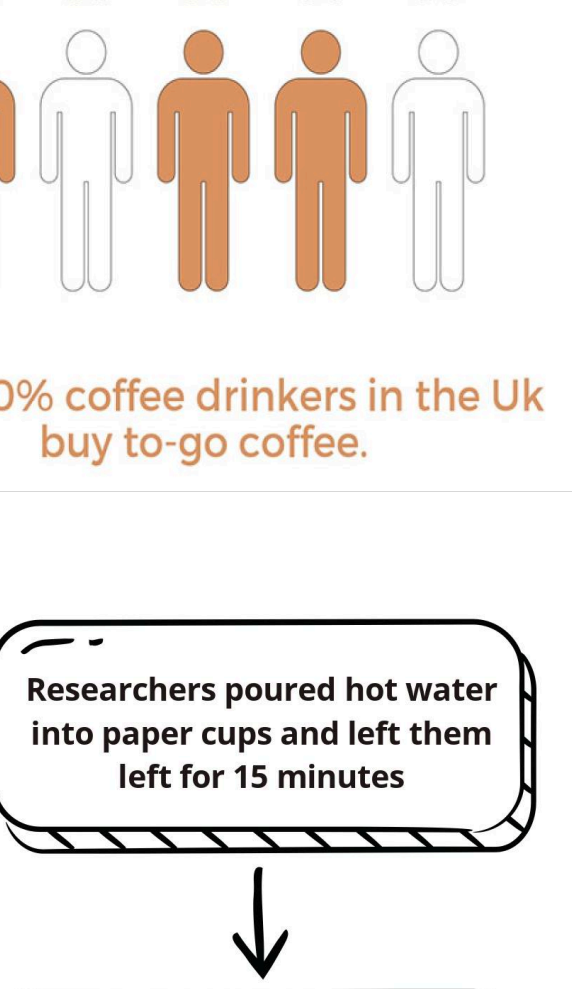
Over 60% coffee drinkers in the Uk buy to-go coffee.

It's a 2 fold problem:

Are the "paper" cups really paper? Seems it is a bit of wax and plastic in it. This makes the recycling process very extensive, complex, costly and labour-intensive. It just ends up utilising more fuel and resources which can be avoided with minor lifestyle changes and enhanced cuppa experience.

Why is there a need for change other than saving the environment? They are Carcinogenic!

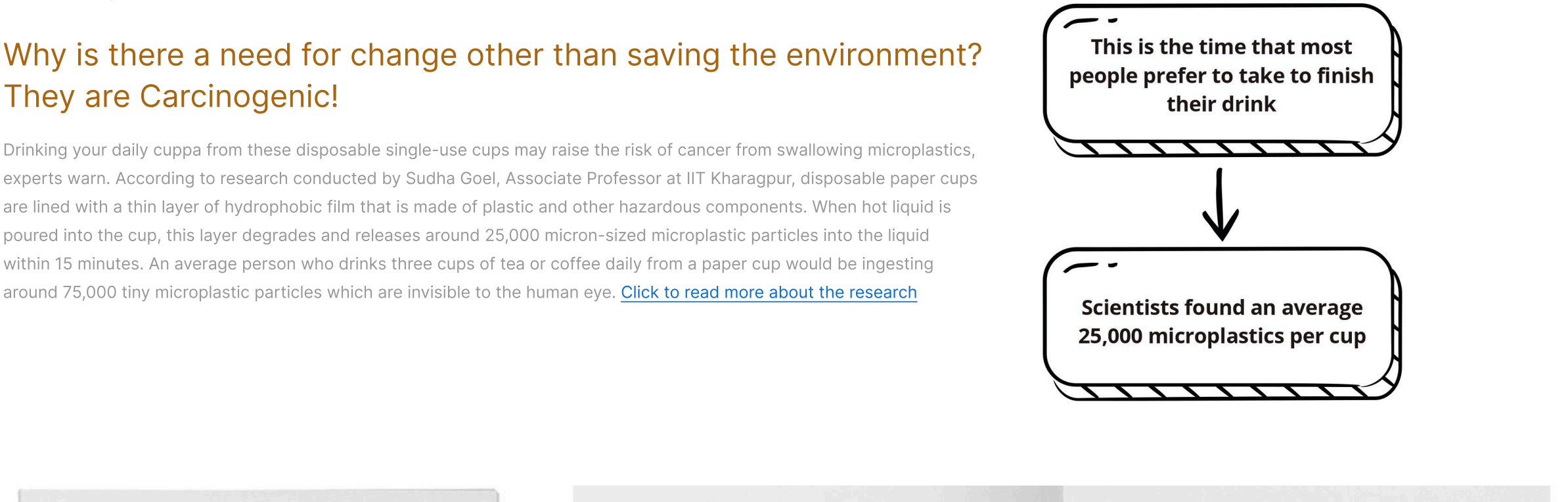
Drinking your daily cuppa from these disposable single-use cups may raise the risk of cancer from swallowing microplastics, experts warn. According to research conducted by Sudha Goel, Associate Professor at IIT Kharagpur, disposable paper cups are lined with a thin layer of hydrophobic film that is made of plastic and other hazardous components. When hot liquid is poured into the cup, this layer degrades and releases around 25,000 micron-sized microplastic particles into the liquid within 15 minutes. An average person who drinks three cups of tea or coffee daily from a paper cup would be ingesting around 75,000 tiny microplastic particles which are invisible to the human eye. Click to read more about the research



Researchers poured hot water into paper cups and left them left for 15 minutes

This is the time that most people prefer to take to finish their drink

Scientists found an average 25,000 microplastics per cup



2

Fold Problem.

Takeaway cups are made from a combination of plastic and paper. Lids pose another problem.

1

Coating with a thin layer of wax to make them water proof and leakproof.

2

Reflecting on research: Ethics, research, and coffee cup waste in the UK

From a design research perspective, many ethical considerations need to be taken into account when addressing the issue of coffee cup waste. These include the following:

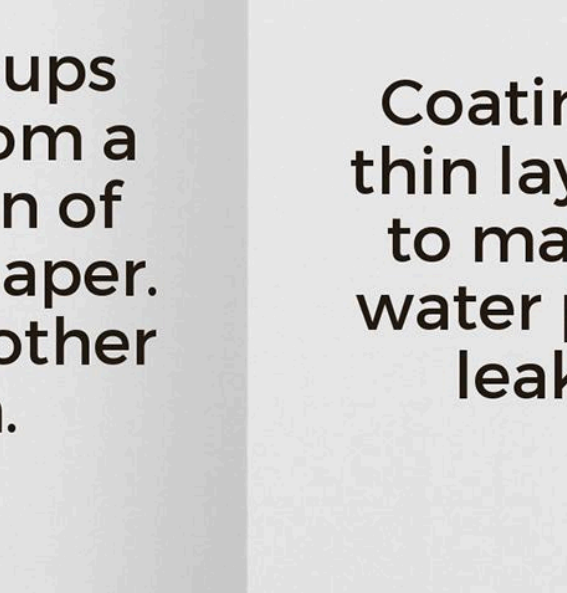
The impact of disposable coffee cups on the environment.

The impact of coffee waste on water resources.

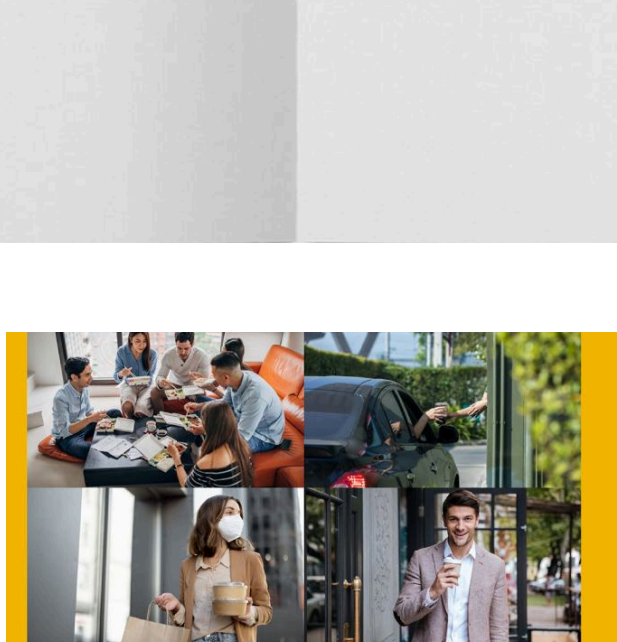
The ethical implications of the coffee industry.

Design interventions that are culturally appropriate.

Evaluate the effectiveness of interventions.



This Cultural Shift.

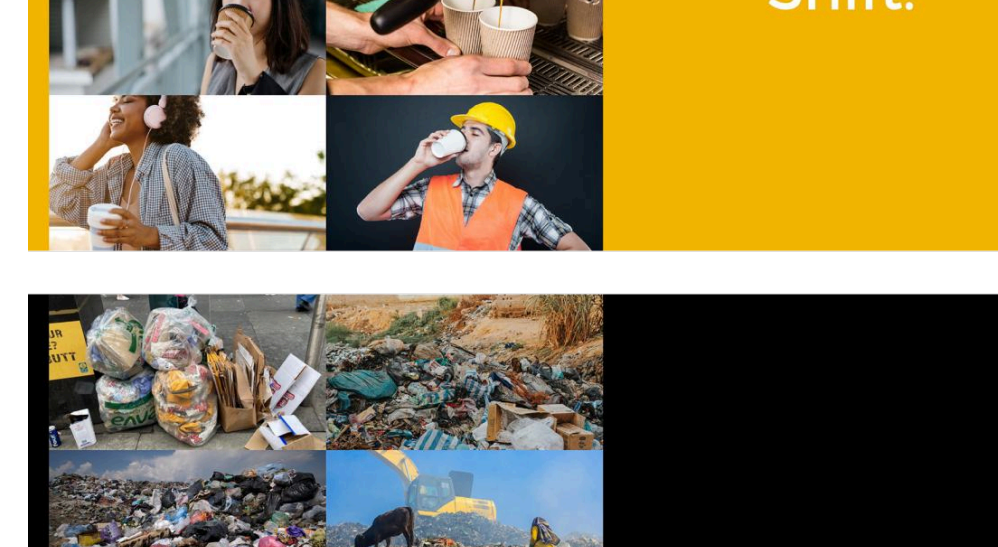


Darker Consequences.

Human Behavior Analysis:

By comprehending the thought process behind a consumer's product selection, marketers can identify opportunities in the market, pinpoint necessary service modifications, and recognize essential touchpoints. The interaction between these human behaviours and increasing environmental consciousness drives the coffee industry to evolve and offer more sustainable solutions to meet consumer demands. The main identified insights are:

Human Behaviour Analysis

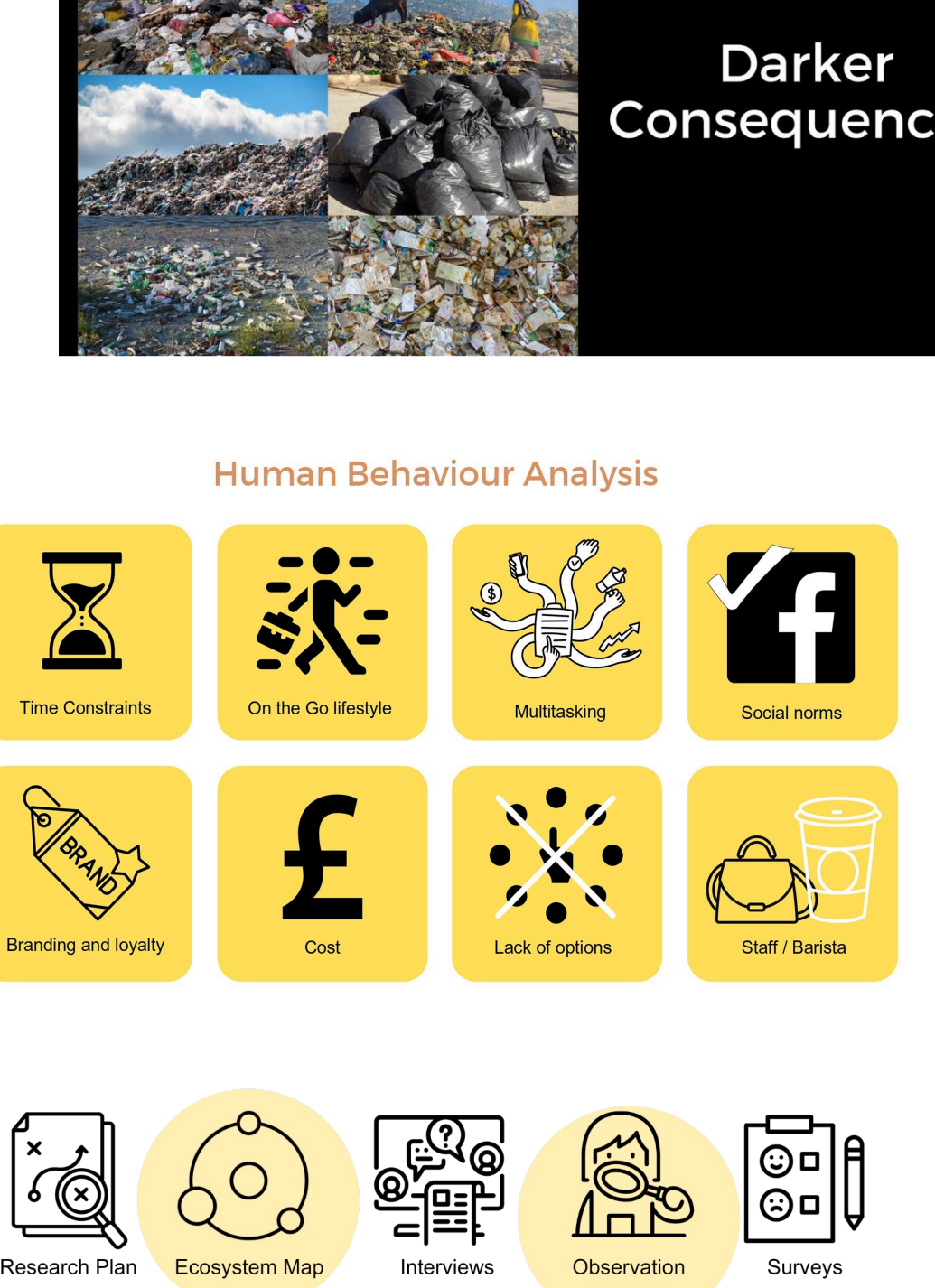

Time Constraints
On the Go lifestyle
Multitasking
Social norms
Branding and loyalty
Cost
Lack of options
Staff / Barista

Engagement Methods: Interviews, Workshops, Observation Sessions, Surveys

Attaining primary information for this project required using different methodologies & tools. To handle the inquiry diplomatically, it's important to avoid reprimanding people for their choices such as using disposable coffee cups. Both the designer and participant benefit from using engagement tools like surveys and interactive installations to foster meaningful conversations.

The approach aimed to connect with coffee lovers, build trust, and address coffee cup waste. By using engagement tools, surveys, and abstraction, the complex issue is broken down into manageable components.

All tools and design research methodologies I used are depicted in the illustration on the right.

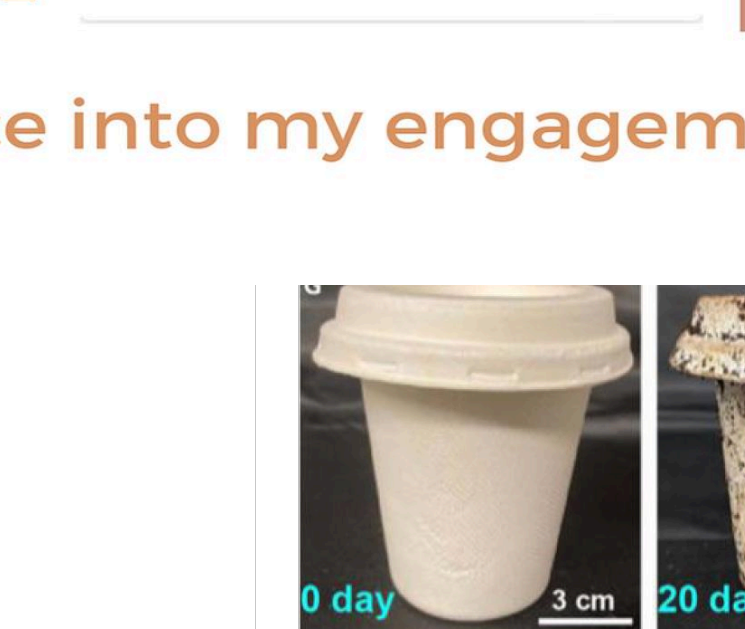

Research Plan
Ecosystem Map
Interviews
Observation Method (photo+video)
Surveys
Assume a Beginner's Mindset
Observational Notes
Bodystorming
Engage with Extreme Users
5 Whys Method
What-How-Why Method
Capture + Share Stories
Synthesis wall
Personal Photo & Video Journals
Mind Map
Personas/ Archetypes
Journey Mapping
Embrace Analogies
Stakeholder Map
Service Blueprint
Emotional Journey
Customer journey

Take a peek into my project

Master's Thesis : Coffee Cup waste

Researcher: Tanvi Kumar
M.Des Design Innovation and Service Design

THE GLASGOW SCHOOL OF ART



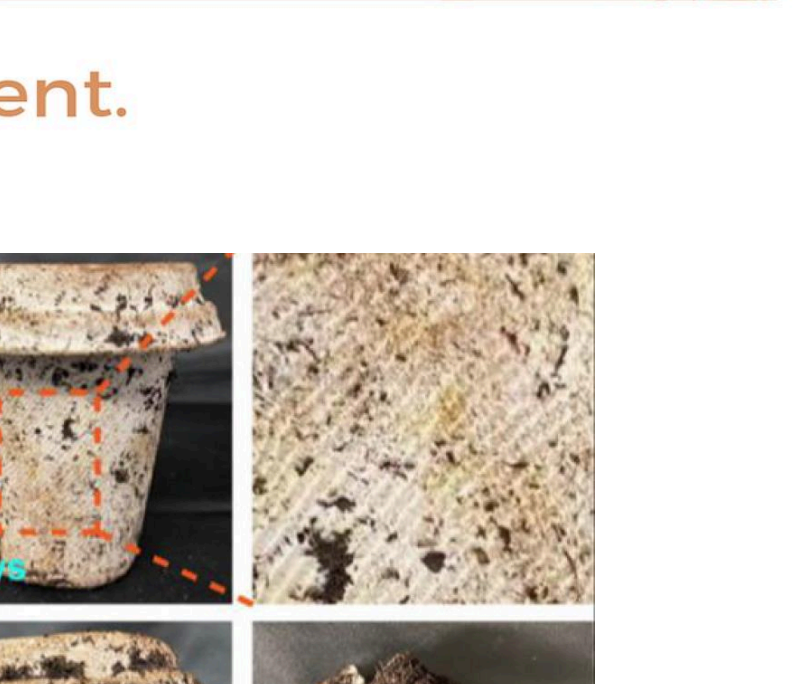
16→ Do you want to share anything else about your opinion on takeaway coffee cups?

Description (optional)

Type your answer here...

Shift ⇧ + Enter ↵ to make a line break

OK ✓ press Enter ↵



Coffee Cups

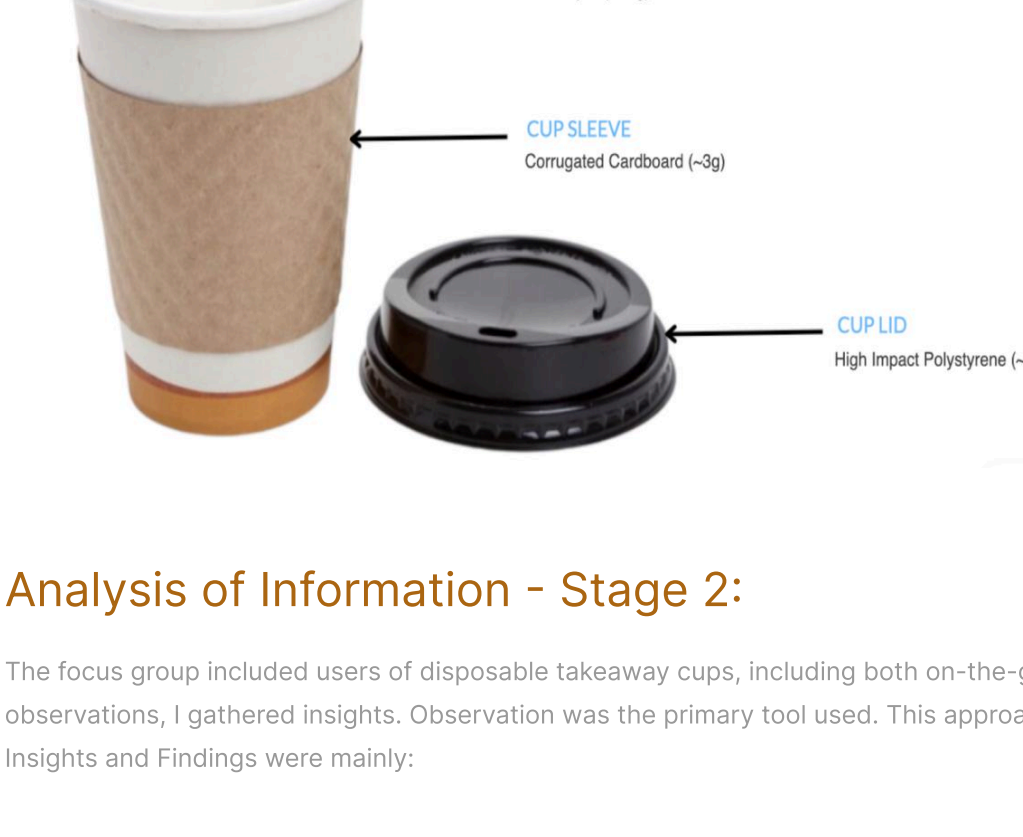
THESIS PROJECT
M.DES SERVICE DESIGN

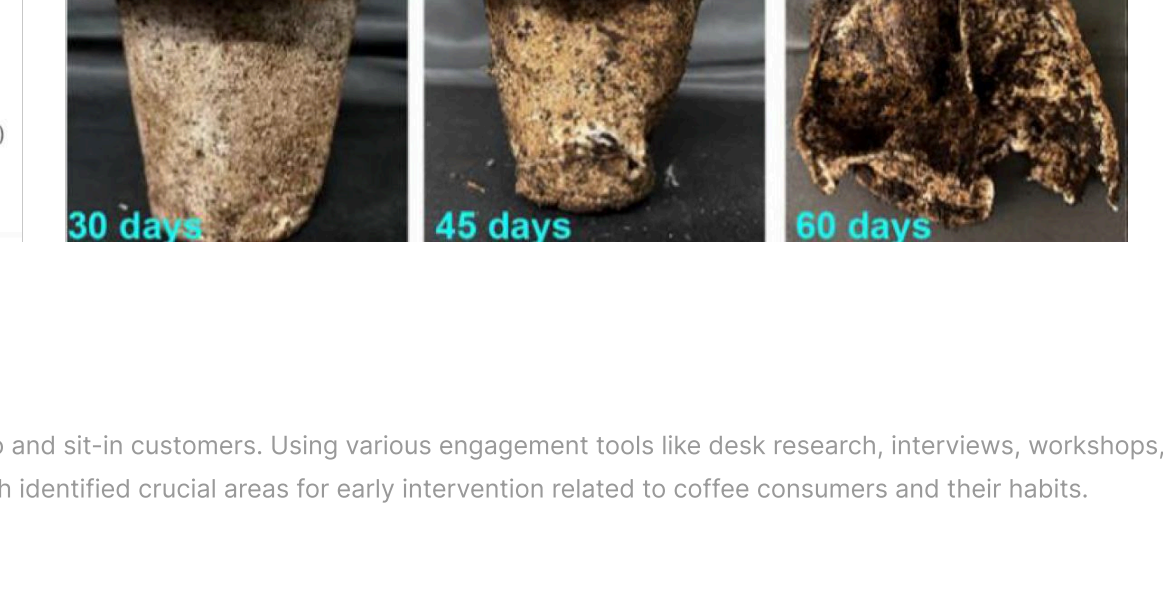
Hey there! I'm Tanvi, a Master's student at the Glasgow School of Art, and I'm currently working on my thesis about coffee cup waste in the UK. To gather accurate data, I'm conducting a survey, and I would absolutely love it if you could spare a few minutes to participate. Your contribution would mean the world to me and my study. It won't take much of your time, but it would make a huge difference. Thanks a bunch in advance!

TAKE THE SURVEY

For any queries or information please contact:
t.kumar1@student.gsa.ac.uk

Glance into my engagement.


INTERNAL WATERPROOF LINING (PLASTIC)
Polyethylene (~01g)
CUP BODY (PAPER)
Paper (~12g)
CUP SLEEVE
Corrugated Cardboard (~3g)
CUP LID
High Impact Polystyrene (~3g)


0 day
3 cm
20 days
30 days
45 days
60 days

Analysis of Information - Stage 2:

The focus group included users of disposable takeaway cups, including both on-the-go and sit-in customers. Using various engagement tools like desk research, interviews, workshops, and observations, I gathered insights. Observation was the primary tool used. This approach identified crucial areas for early intervention related to coffee consumers and their habits.
Insights and Findings were mainly:

Sadly, the coffee-drinking act lasts 10-20 minutes on average. then the cup used is disposed of., usually in conventional ways inappropriately.

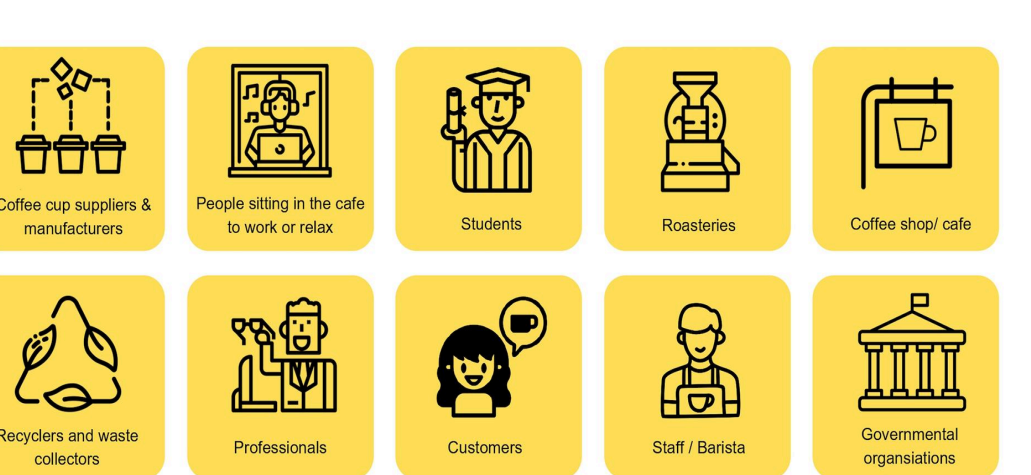
Green buzzwords make compostable or non-plastic cups seem eco-friendly and sustainable. There's still an undeniable timeline for the decomposition and its effects.

The profitable and cheap cups and the high revenue, foot traffic and popularity takeaway customers bring is good reason to continue this vicious cyclic service method.

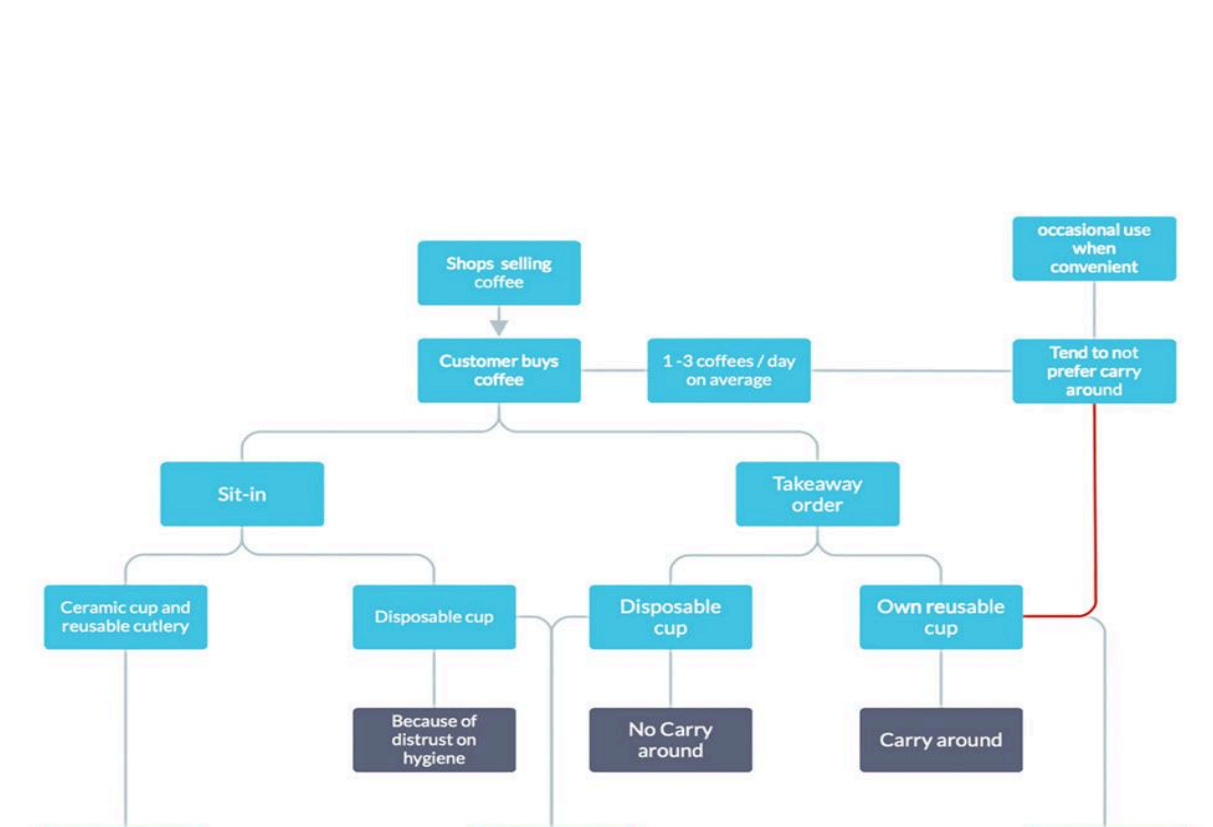
People are so dependent on the service that it is socially acceptable, normal. Willing to pay a little more for the convenience is 'fine'.

Development of Solution:

The solution to address the challenges of takeaway culture in coffee shops and drive-thrus is Crush Cup, a sustainable intervention that focuses on individual- level-consumer-based drivers of behavior change. This solution involves the revival of the terracotta, which would have benefits at various levels.


Coffee cup suppliers & manufacturers
People sitting in the cafe to work or relax
Students
Roasteries
Coffee shop/ cafe
Recyclers and waste collectors
Professionals
Customers
Staff / Barista
Governmental organsiations

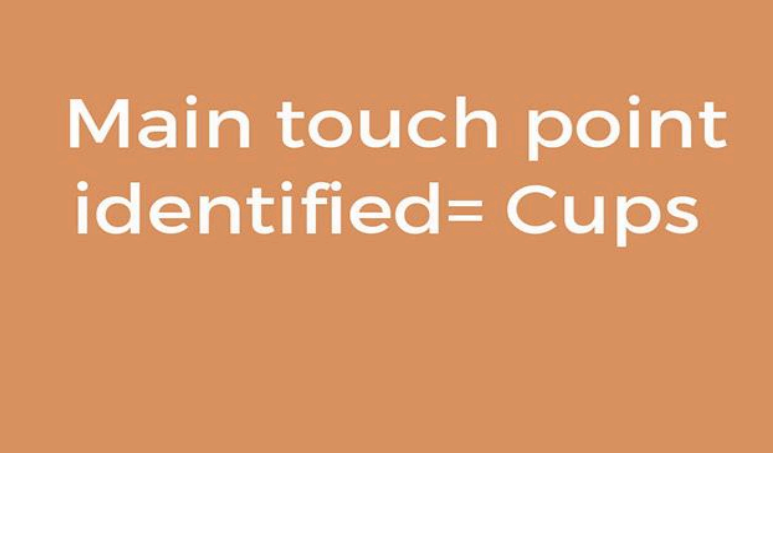
Analysing the relationship between the actors, Insights and problems identified.


Shops selling coffee
Customer buys coffee
1 -3 coffees / day on average
occasional use when convenient
Tend to not prefer carry around
Sit-in
Takeaway order
Ceramic cup and reusable cutlery
Disposable cup
Because of distrust on hygiene
Disposable cup
No Carry around
Own reusable cup
Carry around
No waste
Waste
No Waste
Resources and fuel used at alarming
Only 1% is actually recycled
High costs!
Pollution
Needs Intervention
Sustainability > Convenience
Awareness and Education

Service Flow currently.

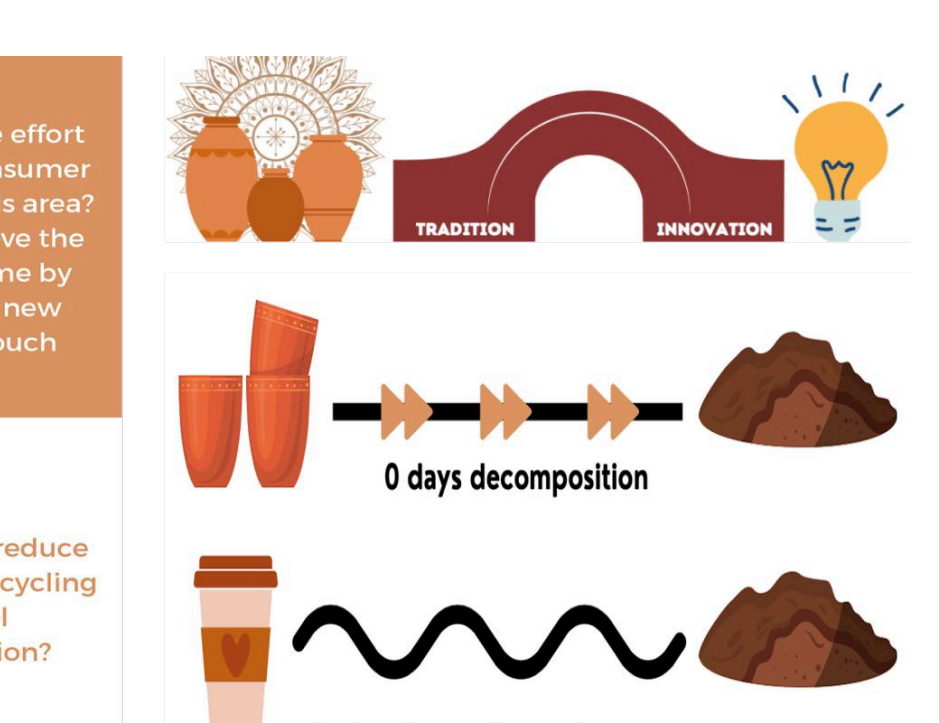
Main touch point identified= Cups

How do consumers in different socio-economic settings dispose of their coffee cups, and what factors influence their choices?	How can a circular economy approach be integrated into the service design to promote the. reuse / recycling of coffee cups and their materials?	How can we make it easier and more convenient for customers to choose sustainable cup options?	Is it worth the effort to change consumer behavior in this area? Or can I achieve the same outcome by designing a new service or touch point?
Can the gap between disposal and decomposition be reduced by the help of a new system design or touch point of the system?	What are the various types of coffee consumers? What are reasons for their behaviour? Which is my niche target group?	What service design interventions can be implemented to incentivise coffee shops to adopt eco-friendly cup alternatives and recycling practices?	How can we reduce the costs of recycling and fuel consumption?

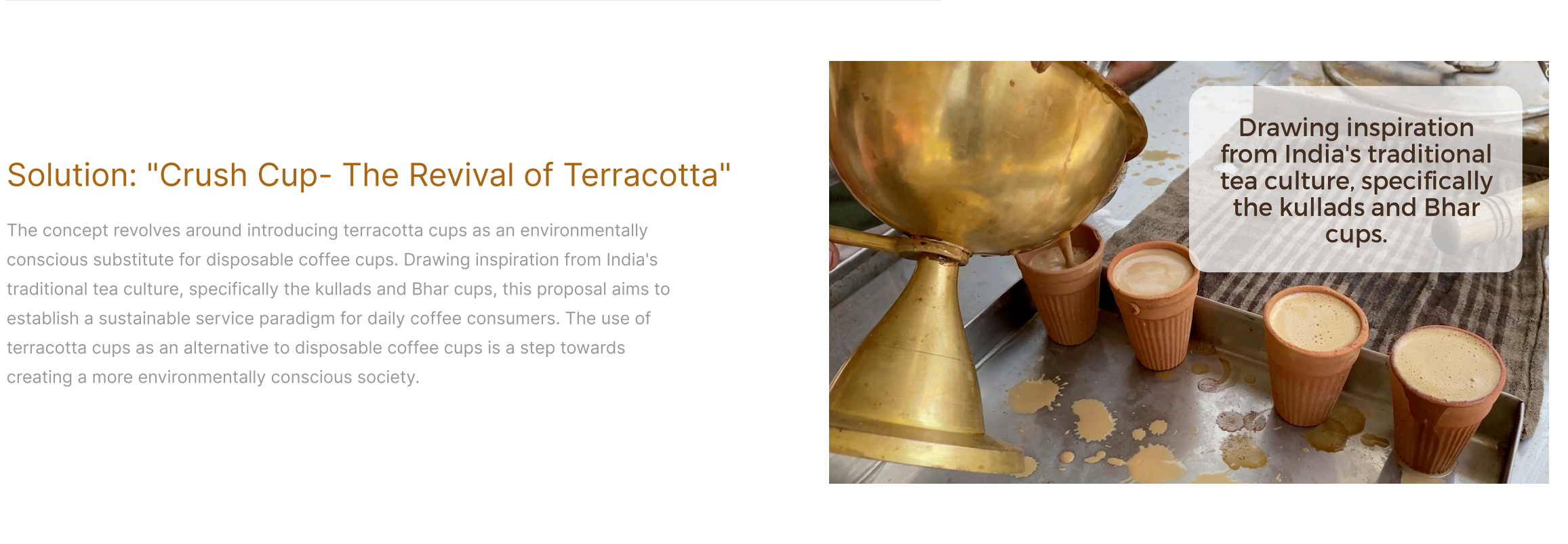

TRADITION
INNOVATION
0 days decomposition
Toxic, slow and complex process

Solution: "Crush Cup- The Revival of Terracotta"

The concept revolves around introducing terracotta cups as an environmentally conscious substitute for disposable coffee cups. Drawing inspiration from India's traditional tea culture, specifically the kullads and Bhar cups, this proposal aims to establish a sustainable service paradigm for daily coffee consumers. The use of terracotta cups as an alternative to disposable coffee cups is a step towards creating a more environmentally conscious society.

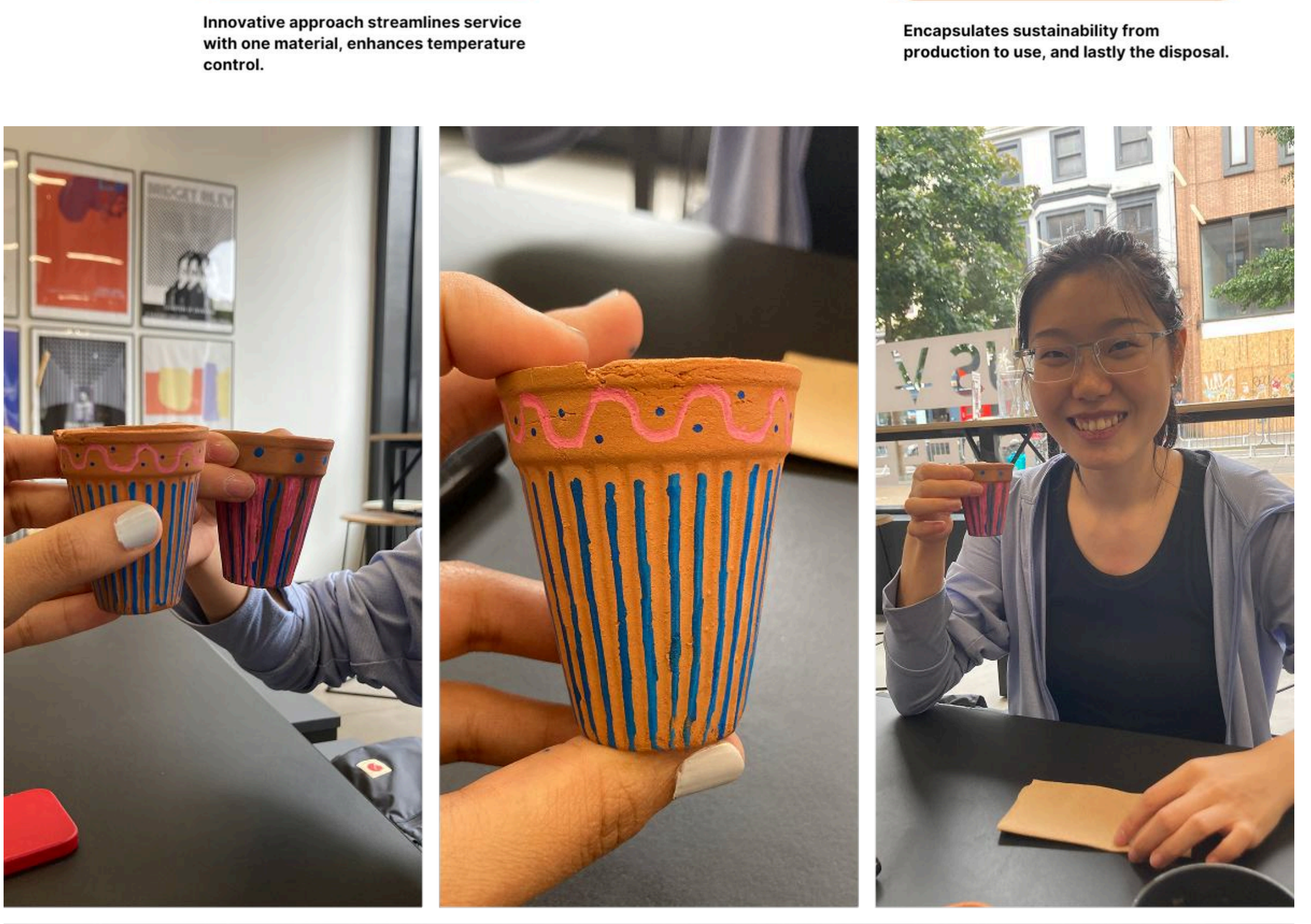


Drawing inspiration from India's traditional tea culture, specifically the kullads and Bhar cups.



India has a well established pottery handicrafts community for centuries now, but this is where the United Kingdom finds itself at the juncture where technology assumes a pivotal role.

Benefits of the Service



SERVICE BENEFITS

TECHNOLOGY

Leveraging clay 3D printing technology makes the task of mass manufacturing in varying designs possible.

EASY DISPOSAL

Disposal options include garden composting or designated bins, Ceremonial crushing on the ground, or planting a sapling in it. promoting eco-friendly practices.

INDUSTRY NEEDS

Cups can be customized to meet industry standards, accommodating hot or cold beverages, takeout, and dine-in preferences.

INFUSING COFFEE

Infusing coffee essence involves repurposing used coffee grounds, dried and finely blended into the clay matrix.

ZERO DECOMP

Zero timeline for the decomposition process. It is made out of mud and becomes mud when disposed of in the environment.

GUILT FREE

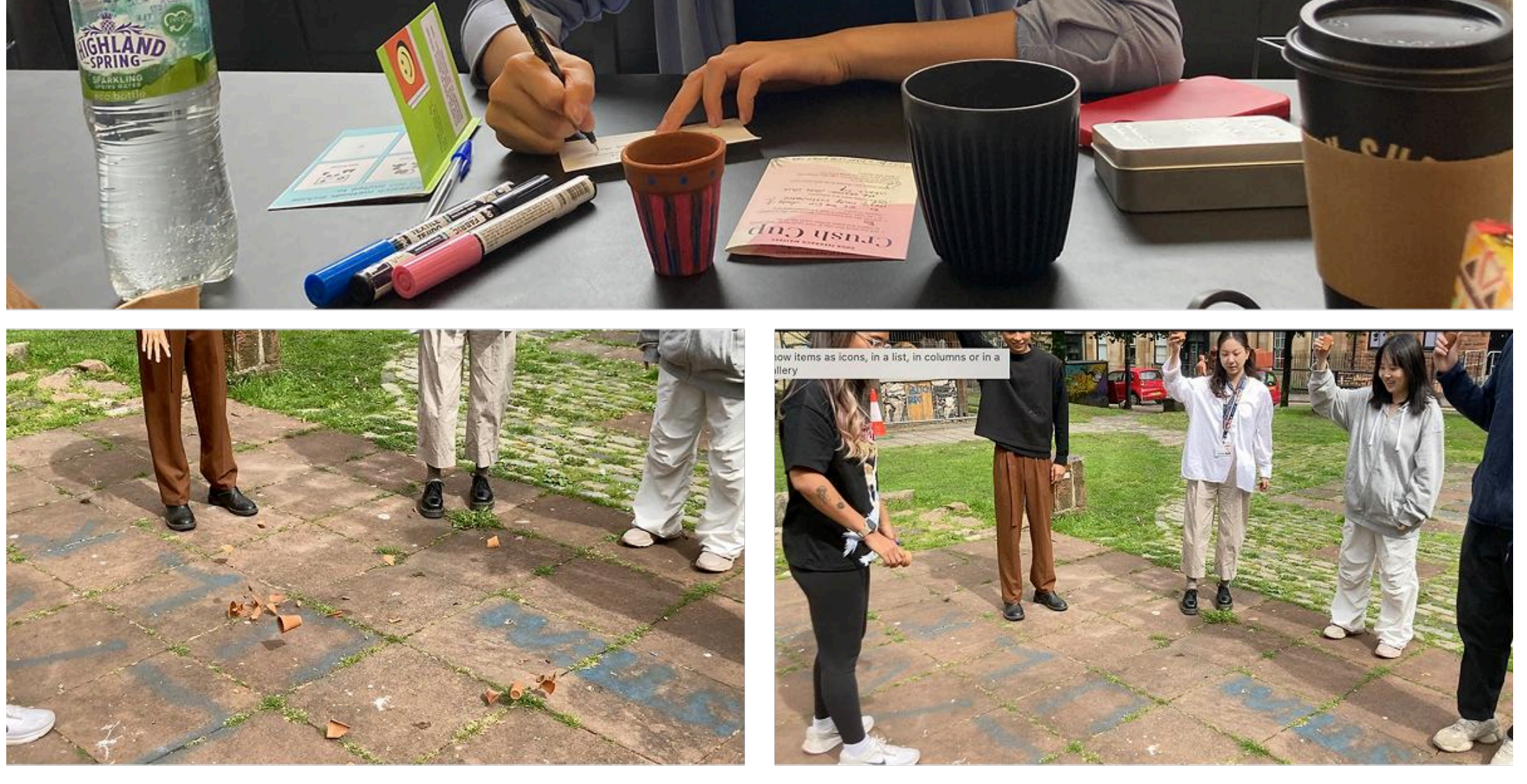
A better emotional journey with the service with no guilt about contributing to the environmental waste of coffee ups, with no major change sin your coffee routine or extra cost.

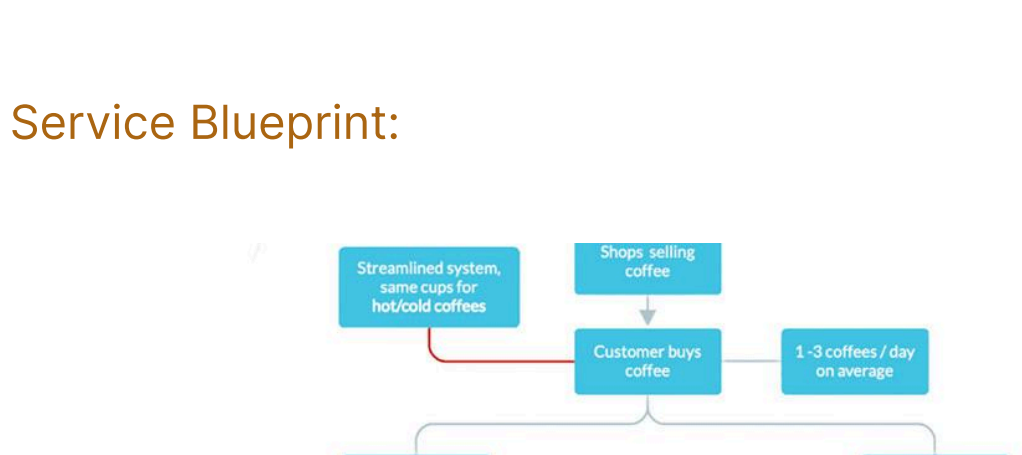
STREMLINES SERVICE

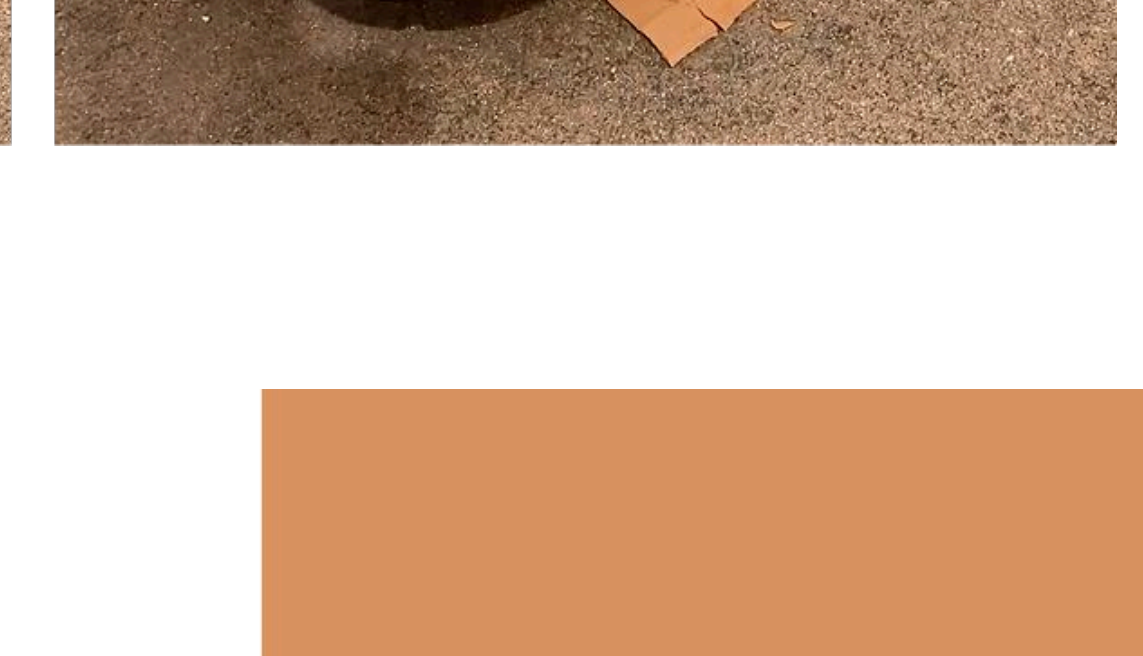
Innovative approach streamlines service with one material, enhances temperature control.

100% SUSTAINABLE

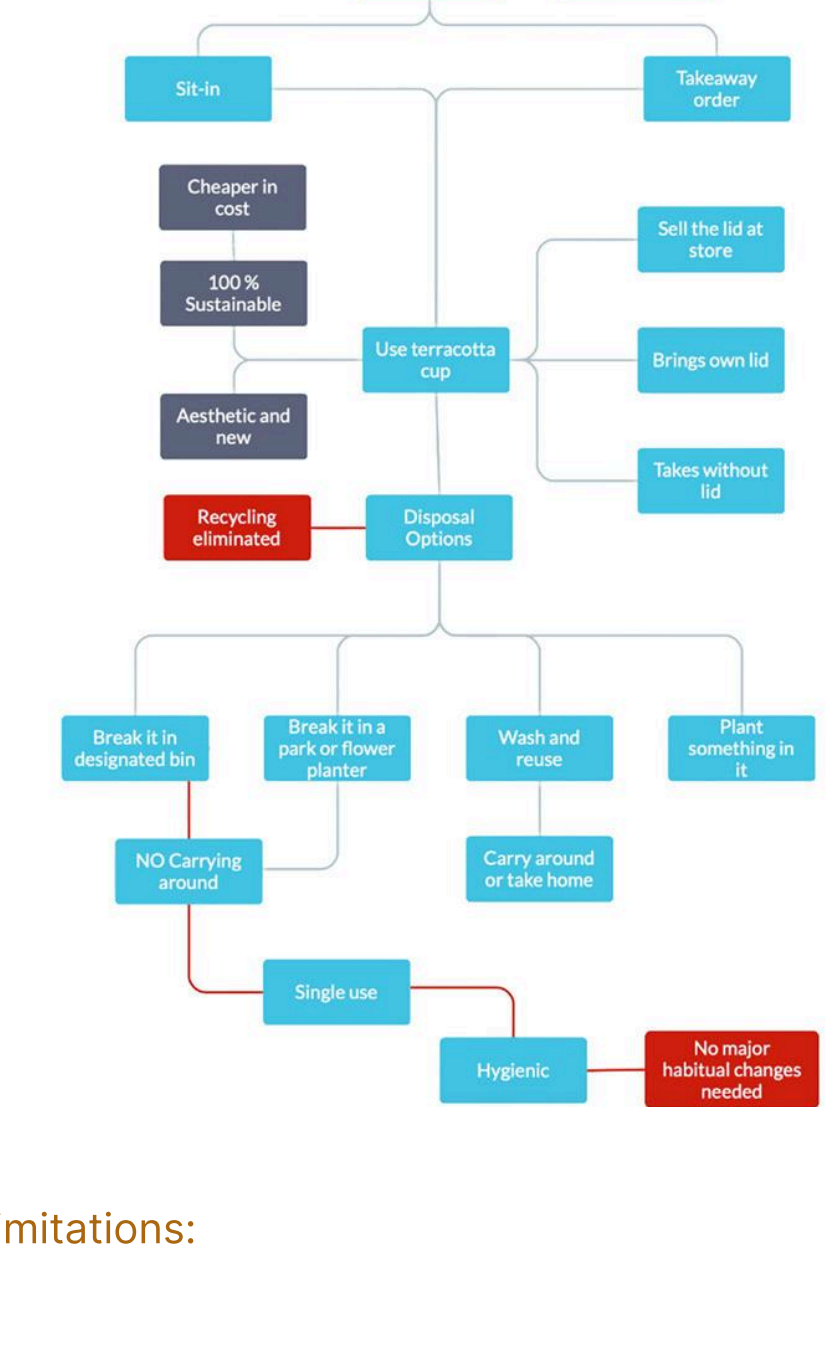
Encapsulates sustainability from production to use, and lastly the disposal.







Service Blueprint:


Streamlined system, same cups for hot/cold coffees
Shops selling coffee
Customer buys coffee
1 -3 coffees / day on average
Sit-in
Takeaway order
Cheaper in cost
100 % Sustainable
Aesthetic and new
Use terracotta cup
Sell the lid at store
Brings own lid
Takes without lid
Recycling eliminated
Disposal Options
Break it in designated bin
Break it in a park or flower planter
Wash and reuse
Plant something in it
NO Carrying around
Carry around or take home
Single use
Hygienic
No major habitual changes needed

Service Flow with Crush Cup.

Possible Limitations:

Marketing	Taste	Cost
Logistics	Cultural Acceptance	Production Time